

Arnab Das

Indian Institute of Science Education and Research, Kolkata.
9064796539 — ad23ms054@iiserkol.ac.in

Objective

Greatly inspired and passionate to work in the field of Astrophysics.

Interests:

- General Relativity and Cosmology.
- Computational Fluid Dynamics.
- Magnetohydrodynamics(MHD).
- Accretion physics.
- Galaxies and Circumgalactic Medium.

Education

ADAMAS INTERNATIONAL SCHOOL

ICSE(Class 10th)
98.60%

Central Model School, Barrackpore

CBSE(Class XII)
95.80%

Indian Institute of Science Education and Research, Kolkata

BS MS Dual Degree
CGPA: 9.62 (Till 4th Semester)

Examination

National Testing Agency(NTA)

JEE(Main) 2023

Score:(236/300). All India Rank(CRL):2004 (99.83 percentile).
100 percentile in Physics.

JEE(Advanced) 2023 conducted by IIT Guwahati.

Score:143/360. All India Rank-6463 (CRL,out of 1.6 lakh students approx.))

West Bengal Joint Entrance Examination 2023

139.5/200 (General Merit List Rank-46)

IISER Aptitude test 2023

171/240 (All India Rank 351)

Experience

- **Introductory School on Astronomy (IISER Mohali):**

This programme consisted of a series of lectures conducted online which covered topics like Coordinate Systems, Roche Limit, Lagrange points, Stellar Evolution and Nucleosynthesis, Gravitational Lensing and many more, including Hands on Sessions.

- **HBCSE,TIFR Mumbai:**

National Initiative for Undergraduate Science (NIUS) Physics Camp Attendee.(Among the top 60 students selected from all over the country) The camp had lectures and short courses on various topics in theoretical and experimental physics, physics education research and an elaborate laboratory course based on experimental problem solving approach.

[Certificate](#)

- **2024 Qiskit fall fest (Conducted by IBM):**

Participated in a Quantum event, and completed a quantum project,as a part of 2024 Qiskit fall fest.

[Winner Certificate:](#)

- **Detecting Dark Matter:** Successfully participating in the “Detecting Dark Matter” workshop organized by Singularity on March 22, 2025 and completing all the given assignments with dedication.

[Certificate](#)

- **Raman Research Institute, Bengaluru:
RRI Astronomy Summer School (RASS), 2025**

I have participated in the Introductory lectures on Astronomy and Astrophysics in RRI Astronomy Summer School on 19-30th May, 2025.

The school consisted of lectures covering a wide range of topics in astronomy and astrophysics:

- Introduction to Astronomy and Astrophysics
- Radiative Processes
- Special and General Relativity
- Astronomical Instrumentation
- Background Cosmology
- Galactic and Extragalactic Astronomy
- Stellar Physics and Compact Objects
- Exoplanets and Planetary Dynamics
- Planetary Studies (Solar System)
- The Sun

[Certificate](#)

Projects

- **Dissipative dynamics of nuclear fusion.**

Date: 1.7.2024-16.8.2024

Supervisor: Dr.Jhilam Sadhukhan, VECC Kolkata.

In this project , I tried to calculate the Capture Cross-section of nuclear capture process(fusion) using a 4-D Langevin Dynamical Model.

Skills developed:

- LaTeX
- FORTRAN Programming Language.
- PYTHON Programming Language.

[Certificate](#)

- **Effect of the Circumgalactic Medium (CGM) on the galactic UV radiation field:**

Date: 1.6.2025-18.7.2025

Supervisor: Dr. Kartick C. Sarkar, RRI, Bengaluru.

- Massive galaxies contain hot virial (10^6 K) gas around them. The project involves finding out the distribution of hot gas based on virial equilibrium and employ the solution to understand the hot gas observation.

- Most astrophysical sources are tested based on the radiation they emit. The radiation also affects its surrounding. The project involves calculating the local radiation field in the Circumgalactic Medium of the galaxy.

[Certificate:](#)

[Project report](#)

Achievements & Awards

- National Standard Examination in Astronomy(2022): Qualified for INAO.
- National Standard Examination in Chemistry(2022): National Top 1%ile Qualified for INChO.
- Kishore Vaigyanik Protsahan Yojana(KVPY): AIR 932(2022) SA Stream.
- National Talent Search Examination(NTSE) Scholar 2021.
- DST INSPIRE Fellow in BS MS Dual Degree at IISER KOLKATA.

Skills

Python
JAVA
Fortran
Gnuplot
MS Office
LaTeX
Github

Courses

- Mechanics I(A+)
- Electricity and Magnetism (A+)
- CS1101:Introduction to Computer Programming(A+)
- Mechanics II (Special Relativity) (A+)
- Waves and Optics (A+)
- Modern Physics Lab I (A+)
- Quantum mechanics I (A+)
- Thermal Physics I (A)
- Math methods of Physics I (A+)
- Optics Lab (A+)
- Introduction to Computation (A+)
- Analysis I (A+)
- Linear Algebra I (A+)
- Probability and statistics I (A+)
- Analysis II (A+)
- Basic Algebra (Group Theory) (A)
- Probability and Statistics II (A)